

|        |       |                                                                                                                                                                                                                       |          |
|--------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7      | (a)   | Random – it cannot predict which and when a nucleus will decay.<br>Nucleus has a constant probability of decay per unit time.                                                                                         | B1<br>B1 |
|        | (b)   | it is not affected by any external factors such as<br>1. temperature, 2. pressure                                                                                                                                     | B2       |
| (c)(i) | 1     | (from Fig 7.1, it can be deduced $T_{1/2}$ for Co is around 5 years<br>and for Ni is around 90 years.)                                                                                                                | B1       |
|        |       | So activity of Co should be higher at the start. Region P is dominantly due to activity of Co.                                                                                                                        | A1       |
|        |       | Hence $\ln A_{\text{mix}}$ against $t$ graph should give a straight line PQ                                                                                                                                           |          |
|        | 2     | In region RS, most of Co has decayed, so activity is mainly due to Ni                                                                                                                                                 | B1       |
|        | 3     | In region QR, both Co and Ni contribute to the activity<br>Activity $A_{\text{mix}}$ is sum of $A_{\text{C}}$ and $A_{\text{N}}$                                                                                      | B1<br>A1 |
|        |       | Hence $\ln A_{\text{mix}}$ against $t$ graph gives a curve.                                                                                                                                                           |          |
| (ii)   | 1     | grad of PQ = $(7.63 - 8.87) / 10$<br>$= -0.124 \text{ yr}^{-1}$                                                                                                                                                       | B1<br>A1 |
|        | 2     | grad of RS = $(4.77 - 5.08) / 40$<br>$= -0.00775 \text{ yr}^{-1}$                                                                                                                                                     | B1<br>A1 |
|        | (iii) | $\ln A = \ln A_0 - \lambda t$<br>Hence decay constant $\lambda$ = magnitude of gradient = same as above<br>decay constant of cobalt = $0.124 \text{ yr}^{-1}$<br>decay constant of nickel = $0.00775 \text{ yr}^{-1}$ | A1       |
| (iv)   |       | $T_{1/2} = \ln 2 / \lambda$<br>Cobalt: $T_{1/2} = \ln 2 / 0.124 = 5.59 \text{ yr}$                                                                                                                                    | M1<br>A1 |
|        |       | Nickel: $T_{1/2} = \ln 2 / 0.00775 = 89.4 \text{ yr}$                                                                                                                                                                 | A1       |
| (d)    |       | Co has very high initial activity (because of large decay constant at the start) – hence more hazardous)                                                                                                              | B1       |
|        |       | award 1 mk if student mentioned both will be hazardous initially as they are in large dosage                                                                                                                          |          |
|        |       | As half life of Co is short (5.59 years) – activity will be small after that number of years, hence less hazardous.                                                                                                   | B1       |
|        |       | However Ni has a long half-life (about 90 years), it remains hazardous for a long time.                                                                                                                               | B1       |